Vadim Piven, Senior Software Engineer

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Software Engineer with **6+ years** of experience **building cross-platform products** with **Rust**, **C++ 20** (STL, Boost), **Objective-C**, and **Qt/QML**. Led product development from conception to completion, focusing on security, observability, performance and seamless user experience.

Professional Experience

C++ / Query Engine Developer
MongoDB, software company that develops and provides commercial support for the database engine MongoDB
August 2026 - Present

Key technologies: C++.

Rust / Node.js Software Developer
MiLaboratories, a biotech startup producing software for immune repertoire profiling with 8 out of 10 Big Pharma companies among its clients
April 2024 - July 2026

- Developed a data lake for biological data, a domain-specific query engine, and a query language providing efficient access to the data in a manner tailored for bioinformaticians. These technologies provide the foundation for the Table and Chart components of the **Platforma.bio** Electron app. The library was initially developed as a WASM-module (using Emscripten), then rewritten into a Node.js Native Addon for better performance, and finally ported to Rust (using **neon-bindings**) to utilize **Apache DataFusion** query engine as an execution backend.
- Led a six-month migration of the entire data-processing stack from a legacy binary format to **Apache Parquet**. This enabled to implement integration with **Polars** (Python) for backend batch data processing and unify the API for data processing across backend and desktop contexts. These changes tripled data processing speed and significantly improved the developer experience for SDK users.

Key technologies: Rust, C++, Node.js (Node-API), WebAssembly, TypeScript, Vite, Electron, Python.

C++ / Qt Software Developer
Movavi, a company focused on products for multi-platform video editing with 3M users in 200+ countries
November 2022 - March 2024

- Created lightweight web-installer with native UI and a curl-based HTTP downloader for both Windows and macOS, which increased successful installation rate by 10% (currently used for distributing all of company products).
- Designed and developed a solution for the marketing team to display advertising banners in products, intended to increase trial-to-paid user conversion rate and estimated to bring additional $6 million revenue a year: product statistics tracker to detect usage patterns, a UI kit for assembling Qt QML banners from pre-designed blocks, utility for screenshotting the banners (distributed in a Docker container), and an agent (to be bundled with all Windows and macOS products) that displays the appropriate banners based on the usage patterns detected.

Key technologies: C++, Qt/QML, Objective-C, Cocoa, Direct2D, Docker, Jenkins, Conan, CMake, Sentry.

C++ / Win32 Software Developer
Kaspersky, an industry leader in endpoint protection, recognized by Gartner as a 2021 Customers' Choice for Endpoint Protection Platforms, 400M individual users and 240K corporate users in 200+ countries
February 2020 - October 2022

- Worked in a team developing the business-logic part of **Kaspersky Endpoint Security** (KES) for Windows (Kaspersky's flagship B2B product for Windows endpoint protection) in accordance with the Microsoft Security Development Lifecycle.
- Introduced Windows Enterprise Multi-Session support by enforcing proper thread impersonation, which allowed to use KES in the Azure Virtual Desktop environment and target new customers.

- Implemented queueing of protection tasks to prevent high disk and CPU utilization during user activity periods which resolved performance problems for customers with the outdated hardware.
- Integrated **EDR** component (MSI installation option, settings migration, upgrade without reboot, local C#/WPF UI settings page, remote MMC-based administration console settings page, security report propagation to local SQLite-based storage and remote administration console storage, network-provided security policy enforcement, fully covered with unit-tests, major scenarios covered with E2E automated tests) to introduce KES to EDR solutions market.

Key technologies: C++, C#/WPF, WiX Toolset, SQLite, WinAPI, WinDbg, Windows Performance Analyzer.

Education

M.S. in **System and Software Engineering**, GPA: 7.25 out of 10
HSE University, consistently ranked as one of Russia's top universities, ranked 151-200 by QS: World University Ranking in Computer Science & Information Systems category
September 2020 - July 2022

B.S. in **Applied Mathematics and Informatics**, GPA: 4.93 out of 5
MIREA - Russian Technological University, one of the leading Russian universities in information and computer technologies, ranked 26 by Best Global Universities in Russia
September 2016 - July 2020